ENVS 193DS Homework 03

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Table of contents


```
# reading in packages
library(tidyverse)
library(here)
```

```r
library(janitor)
library(readxl)
library(scales)
# creating new object from kelp data
kelp <- read_csv((here::here ("data","temp-kelp.csv")))
# creating new object from personal data
personal_data <- read_csv((here::here ("data","personal-data.csv"))) |>
  # cleaning names of columns
  clean_names() |>
  # dropping N/A values
  drop_na(date)
```

Problem 1: Giant Kelp fronds

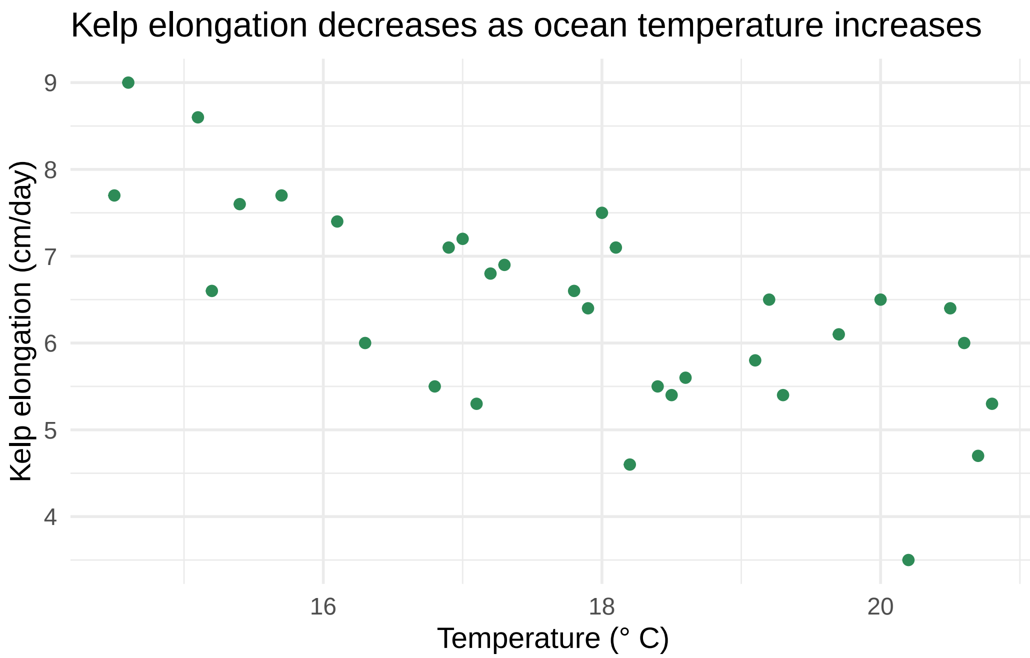
(a)

One method of determining the strength of a relationship between temperature and giant kelp frond elongation is by finding the Pearson's correlation (r), which determines the direction and strength of the linear relationship between variables. The non parametric version is the Spearman rank correlation, which determines the monotonic relationship between variables.

(b)

```r
# base layer ggplot
# data: kelp
ggplot(data = kelp,
       mapping = aes(
         # x-axis: temperature
       x = temp_c,
        # y-axis: kelp elongation
       y = kelp_elong)) +
  # first layer: scatterplot
  # changing color to royal blue
  geom_point(col = "seagreen") +
  # changing x and y axis titles, adding title
  labs(x = "Temperature (\u00B0 C)",
       y = "Kelp elongation (cm/day)",
       title = "Kelp elongation decreases as ocean temperature increases") +
```

```
# changing to minimal theme
theme_minimal()
```



(c)

Checking assumptions:

In part 1b, I created a scatterplot to observe the patterns displayed in the `kelp` data. In this plot, a negative linear relationship between ocean temperature and kelp elongation is visible. This indicates that finding the Pearson's correlation would be the best way to determine the strength of this relationship since linearity is an assumption of this test. The other assumptions are met since the data is continuous and we can assume that the observations are independent.

Running a test

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
         # data from kelp dataframe
         data = kelp,
```

```
         # pearson correlation
         method = "pearson")
```

```

    Pearson's product-moment correlation

data:  temp_c and kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

(d)

To evaluate the strength of the relationship between ocean water temperature and kelp elongation, I used a Pearson's correlation test because the data is linear. I found a moderate negative (Pearson's r = -0.687, t(30) = -5.19, p = 1.366e-05, 03B1 = 0.05) relationship between ocean temperature and kelp elongation. This leads me to reject the null hypothesis that the correlation is equal to 0.

(e)

I found that kelp elongation rates tend to decrease as temperatures increase. There is a moderate negative relationship between these two variables (Pearson's r = -0.687, p = 1.366e-05, 03B1 = 0.05). In a time of rising temperatures worldwide, this is important to know as giant kelp is the base for many marine ecosystems.

(f)

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
         # data from kelp dataframe
         data = kelp,
         # spearman correlation
         method = "spearman")
```

```
    Spearman's rank correlation rho

data:  temp_c and kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

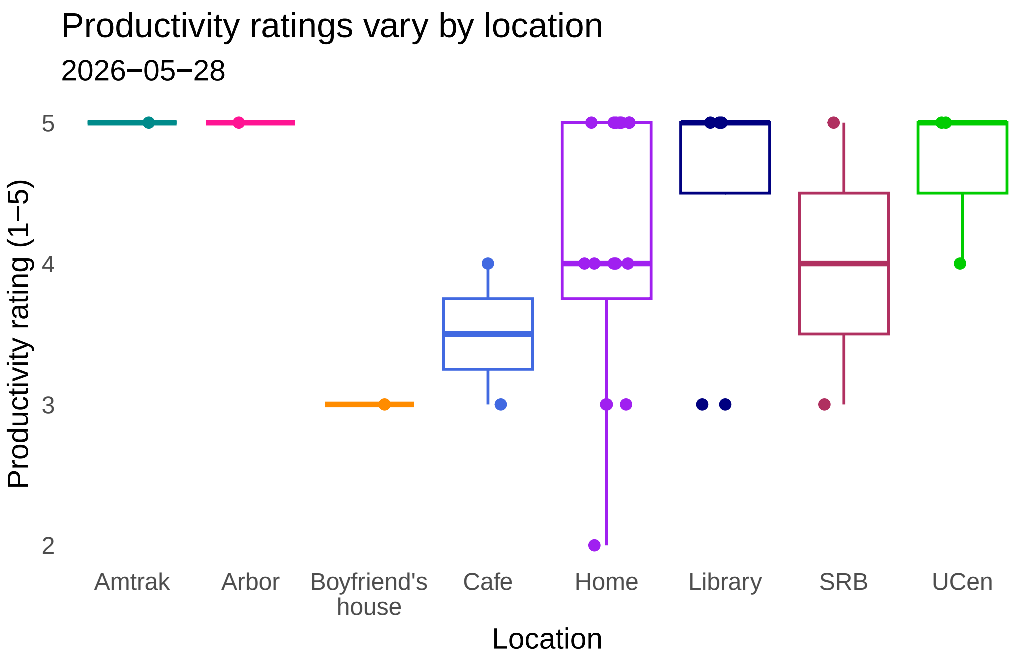
These two tests have led me to the same interpretation of results. In the Spearman correlation test, I found a moderate negative relationship (Spearmans $\rho = 0.689$, S = 9216.1, p = 1.29e-05, 03B1 = 0.05) between temperature and kelp elongation. This information leads me to reject the null hypothesis that rho is equal to 0.

Problem 2: Personal Data

(a)

```
# ggplot base layer, data from personal_data
ggplot(data = personal_data,
       # x-axis: location
       # y-axis: productivity rating
       mapping = aes(x = location,
                     y = productivity_rating_1_5,
                     # color by location
                     color = location)) +
  # boxplot layer 2
  geom_boxplot() +
  # jitter plot layer 3
  geom_jitter(height = 0, width = 0.2) +
  # changing titles of axes, adding title and subtitle
  labs(x = "Location",
       y = "Productivity rating (1-5)",
       title = "Productivity ratings vary by location",
       subtitle = "2026-05-28") +
  # removing grid, making background blank
  theme( panel.background = element_blank(),
    panel.grid = element_blank(),
    axis.ticks = element_blank()) +
```

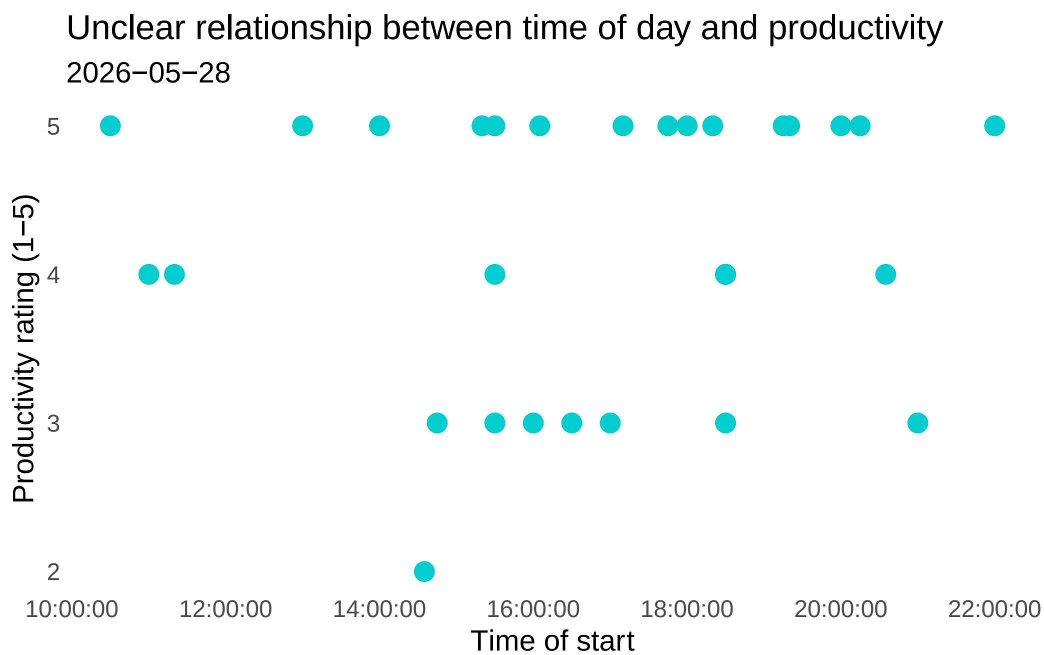
```
  # changing colors of graph
  scale_color_manual(values = c(
    "darkcyan", "deeppink", "darkorange", "royalblue", "purple", "navy", "maroon", "green3"))
  # wrapping location titles to avoid overlap
    scale_x_discrete(labels = label_wrap(10)) +
  # removing legend
  theme(legend.position = "none")
```



add subtitle, wrap x-axis labels?, take out grids and background

```
# ggplot base layer, data is personal_data
ggplot(data = personal_data,
       # x-axis: time of start
       # y-axis: productivity rating
       mapping = aes(x = time_of_start,
                     y = productivity_rating_1_5)) +
  # add points
  geom_point(
    # change color and size of points
    size = 3, color = "cyan3") +
  # adding labels to axes and adding title
  labs(x = "Time of start",
       y = "Productivity rating (1-5)",
```

```
    title = "Unclear relationship between time of day and productivity",
    subtitle = "2026-05-28") +
# apply minimal theme
# removing grid, making background blank
theme( panel.background = element_blank(),
  panel.grid = element_blank(),
  axis.ticks = element_blank()) +
# remove legend
theme(legend.position = "none")
```



(b)

Figure 1: Comparison of productivity levels at different locations. Each boxplot represents a location where I have studied, and each point represents a single study session.

Figure 2: The relationship between studying start time and productivity. Each point represents a single study session.

Problem 3: Affective visualization

(a)

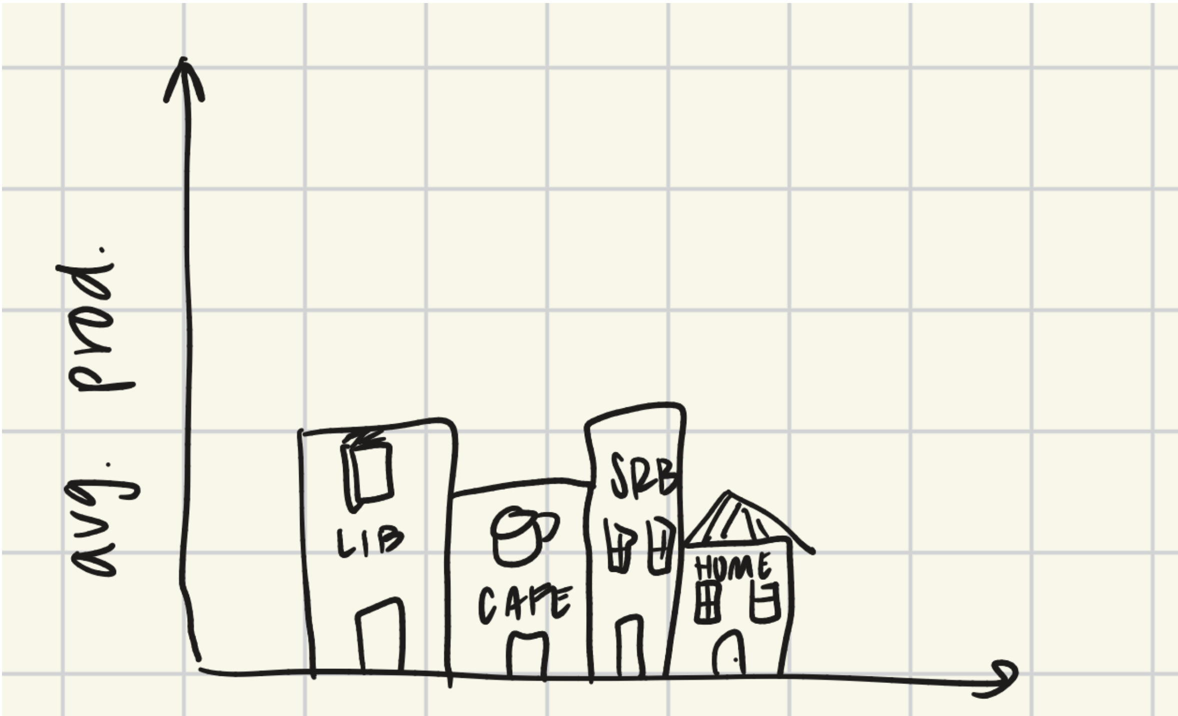
One idea I had for an affective visualization of my data is to create a bar graph where the x-intercept is the different locations, and the y-intercept is the average productivity rating there. I would take this and draw it by hand to make each bar a drawing of a little building that represents where I was studying.

(b)

```
# adding sketch image
knitr::include_graphics((here::here ("photos","3b.png")))
```

(c)

```
# adding sketch image
knitr::include_graphics((here::here ("photos","3c.png")))
```



(d)

This shows my the changes in my productivity when studying when I am at different locations. I was inspired by the Dear Data project by Stefanie Posavec and Giorgia Lupi as well as Jill Pelto's artwork. The draft was done on my ipad, and the final draft will likely be either digital or done with colored pencils, pens, and markers on paper.

(e)

View Slides

Problem 4: Statistical critique

(a)

The test I have selected is the Mann-Whitney U-test. The response variable is treefrog density and the predictor variables are urbanization and crayfish presence. There is not a figure or table that is associated with the Mann-Whitney test that I am focusing on.

(b)

The test I am focusing on does not have a figure or a table.

```r
# adding stream image
knitr::include_graphics((here::here ("photos","stream-screenshot.png")))
```

Table 5. Macroinvertebrate community indices in urban and natural streams in the Santa Monica Mountains and Simi Hills, California.

	2001	
	urban *(n = 13)*	***natural*** *(n = 20)*
Taxonomic richness	23.15[a]	29.40
EPT[b] taxa	5.08[a]	9.40
Percent EPT invertebrates	23.26	32.98
Percent senstive EPT (TV[c]= 0–2)	0.97[d]	13.33
Percent intolerant (TV = 0–3) organisms	1.03[e]	10.65
Percent tolerant (TV = 8–10) organisms	13.34[f]	9.90
Percent most dominant taxon	45.91[e]	33.69
Shannon diversity	1.65[d]	2.23

[a] *Significant difference between urban and rural, low-gradient streams, bonferroni comparisons based on overall* p = *0.05.*

[b] *Aquatic insect orders Ephemeroptera, Plecoptera, and Trichoptera.*

[c] *Tolerance values, a measure of sensitivity to disturbance and pollution with 0 being most sensitive and 10 most tolerant.*

[d] *Significant difference between urban and rural streams at* p < *0.01*

[e] *Significant difference between urban and rural streams at* p < *0.05.*

[f] *Significant difference between urban and rural streams at* p < *0.10.*

This table shows the macro invertebrate community indices in urban and natural streams in the Santa Monica Mountains and Simi Hills, CA in 2001. It displays rows such as taxonomic richness and percent EPT (aquatic insect orders Ephemeroptera, Plecoptera, and Trichoptera) invertebrates, and compares urban and natural streams. Data points marked with different superscripts identify significant differences between the two at different p-values, bonferroni comparison, and other**.